



COURSE DESCRIPTION FORM

AI-2001 Programming for Artificial Intelligence - (PAI)

Institution: FAST School of Computing, National University of Computer and Emerging Sciences, Karachi.

Program: BS AI – Fall 2025

Course Description

Course Code	AI-2001										
Course Title	Programming for Artificial Intelligence										
Credit Hours	3 + 1										
Prerequisites by Course(s) and Topics	NIL										
Grading Policy	Absolute										
Policy About Missed Assessment Items in the Course	Retakes for missed assessment items (excluding midterm and final exams) will not be conducted. In the case of a missed midterm or final exam, an exam re-take or pre-take application, along with the necessary supporting evidence, must be submitted to the department secretary. The Examination Assessment and Retake Committee will review and decide on all exam re-take and pre-take requests.										
Course Plagiarism Policy	Plagiarism in project or midterm/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.										
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	60% Theory, 40% Practical <table> <tr> <th>Assessment Item</th><th>Weight (%)</th></tr> <tr> <td>Assignments</td><td>10</td></tr> <tr> <td>Midterm Exam</td><td>30 (15 + 15)</td></tr> <tr> <td>Quizzes + Project</td><td>10</td></tr> <tr> <td>Final Exam</td><td>50</td></tr> </table>	Assessment Item	Weight (%)	Assignments	10	Midterm Exam	30 (15 + 15)	Quizzes + Project	10	Final Exam	50
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Assignments	10										
Midterm Exam	30 (15 + 15)										
Quizzes + Project	10										
Final Exam	50										
Head of the Department	Dr. Muhammad Rafi										



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Course Instructor	Talha Shahid
Lab Instructor	Muhammad Khalid
Course Coordinator	----
Current Catalog Description	This course aims to introduce standard programming practices and to help develop programming skills necessary for designing and implementing Artificial Intelligence systems. The course introduces a modern state of the art programming language for Artificial Intelligence, and builds up the necessary programming background for the main courses like Knowledge Representation & Reasoning, Machine Learning, Artificial Neural Networks, and Natural Language Processing. This course will help the students of Artificial Intelligence develop the programming acumen and style. The ultimate aim of this course is to help students in using the programming language to solve problems of interest to them.
Textbook (or Laboratory Manual for Laboratory Courses)	1. Severance, C. R., 2016. Python for Everybody: Exploring Data Using Python 3. CreateSpace Independent Publishing Platform, p. 79. 2. Miller, B. N., Ranum, D. L., and Anderson, J., 2019. Python Programming in Context. Jones & Bartlett Learning. 3. McKinney, W., 2012. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython. O'Reilly Media, Inc.
Reference Material	1. Joshi, P., 2017. Artificial Intelligence with Python. Packt Publishing Ltd. 2. Janert, P. K., 2010. Data Analysis with Open-Source Tools: A Hands-on Guide for Programmers and Data Scientists. O'Reilly Media, Inc.
Course Learning Outcomes	A. Course Learning Outcomes (CLOs) On successful completion of this course students will have to know how of:



CLO No.	CLO	Bloom's Taxonomy
01	Comprehend the fundamental construct of programming language for data analysis and representation.	C2 Understand
02	Understand and apply the Object-Oriented concepts in the programming language.	C2 Understand
03	Analyze and solve problems using programming, data analysis and standard libraries and/or toolboxes of the programming language.	C4 Analyze

B. Program Learning Outcomes

For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.

No.	PLO	PLO Detail	
01	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution complex computing problems.	✓
02	Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems to reach substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✓
03	Design/Develop Solutions	Design solutions for complex computing problems and develop	



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			systems, components, and processes that meet specified needs, with appropriate consideration for public health and safety, as well as cultural, societal, and environmental factors.	
	04	Investigation & Experimentation	Conduct investigations of complex computing problems using research-based knowledge and methodologies.	
	05	Modern Tool Usage	Create, select, and apply appropriate techniques, resources, and modern computing tools, including prediction and modeling, to address complex computing problems.	✓
	06	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to the context of complex computing problems.	
	07	Environment & Sustainability	Understand and evaluate the sustainability and impact of professional computing work in solving complex computing problems.	
	08	Ethics	Apply ethical principles and commit to professional ethics, responsibilities, and norms of computing practice.	
	09	Individual & Team Work	Function effectively as an individual and as a member or leader in diverse teams	



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			and multidisciplinary settings.	
	10	Communication	Communicate effectively on complex computing activities with the computing community and society at large.	
	11	Project Management & Finance	Demonstrate knowledge and understanding of management principles and economic decision-making, and apply these to one's own work as an individual or team member.	
	12	Life Long Learning	Recognize the need for, and demonstrate the ability to engage in, independent and lifelong learning in the context of ongoing technological changes.	

Topics covered in the course with number of lectures on each topic (assume 15 weeks of instruction)	Week	Topics
	01	Introduction to Python, setting up the development environment (Jupyter Notebooks). Cover core concepts: variables, expressions, statements, operands, and operators.
	02	Mastering program flow with conditional statements (if/else) and loops (for, while). Introduction to writing efficient, single-line queries and list comprehensions. In-depth exploration of Python's primary data structures: lists, dictionaries, tuples, and sets . Focus on their use cases, methods, and performance characteristics for data handling.
	03	Writing custom functions to create reusable and modular code . Cover arguments, return values, and scope . Introduction to useful built-in functions like zip and enumerate , etc., and anonymous functions using lambda .



	04	Understanding the principles of OOP: classes, objects, inheritance, encapsulation, and polymorphism.
	05	Reading from and writing to local files (e.g., CSV, TXT, JSON). Introduction to interacting with web APIs to fetch live data, handle responses, and parse JSON into Python objects .
	06	Theory Mid-Exam-I
	07	Introduction to the NumPy library for high-performance numerical operations. Focus on creating and manipulating N-dimensional arrays (vectors, matrices, tensors) , array broadcasting, and vectorization.
	08	Introduction to the Pandas library for data analysis. Learn to use Series and DataFrames for data loading, cleaning, filtering, transformation, grouping, and merging.
	09	Lab Mid-Exam
	09	Principles of effective data visualization. Using Matplotlib to create foundational plots (line, bar, scatter, histogram). Introduction to Seaborn for creating more sophisticated statistical graphics.
	10	Machine Learning I
	11	Theory Mid-Exam II
	12	Machine Learning II
	13	NLP Option: Focus on the programming task of preparing text for AI. Covers text cleaning, normalization, and vectorization techniques (e.g., Bag-of-Words, TF-IDF) using libraries as tools.



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	14	CV Option (image processing): Focus on the programming task of handling images. Covers image representation as NumPy arrays, and preprocessing techniques (resizing, color space conversion, normalization) using libraries like OpenCV .		
	15	Writing professional, maintainable code (documentation, modularity). A dedicated lecture on Ethical AI (data privacy, bias).		
	16	Theory Final Exam		
Programming Assignments Done in the Course	All the assignments would be programming based (Python)			
Class Time Spent (in percentage)	Theory (%)	Problem Analysis (%)	Solution Design (%)	Social and Ethical Issues (%)
	50	25	20	5
Oral and Written Communications	Every student is required to submit at least 1 written report of typically 3 pages. Students will also be called for viva/presentation of the project and any assignment where necessary.			

Course Instructor: **Talha Shahid**

Signature:

Dated: **18-Aug-2025**